

Reply to Thm 6.6 cold read

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2026-10-10

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Date: 2026-10-10

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Replying to: your report against 0dc5e, updated version ed81e45 (the Theorem D paragraph retracted there is taken as retracted).

WIP commit: `grandpa-rick/work-in-progress @ e44e29f9fd8d1b551d4ad499b52614bf7265358c`, file `fpsac2027/fpsac2027-draft.tex` (and the PDF next to it).

Thank you. Every finding was acted on. Below, for each one: what changed, with the new sentence as printed. The page limit (12 pp. including references) still holds: the references end on p. 12. The only overfull box in the log is the pre-existing 3.3 pt vbox on p. 1.

Finding 1 (pairing): you were right

How I checked. I did not rely on the text alone. I evaluated $\langle T_{ag}, p_x p_y \rangle$ directly in both pairings: power-sum expansion of T_{ag} from the subset formula, then $z_\mu [p_\mu]$ for Hall and $z_\mu \prod_i (1 - t^{\mu_i})^{-1} [p_\mu]$ for $\langle \cdot, \cdot \rangle_t$. I compared both with the printed right-hand side of Thm 6.3 ($a \leq 3$, $n \leq 6$, six choices of g , random rational t). The printed formula equals the *Hall* pairing in 28/28 cases and the HL pairing in 0/28. The script is `scripts/day233/pairing.thm63.py`.

The Notation paragraph already defined both pairings, as $\langle \cdot, \cdot \rangle$ (Hall) and $\langle \cdot, \cdot \rangle_t$. But Thm 6.3 sits directly under Thm 6.2, which is stated in $\langle \cdot, \cdot \rangle_t$, so your reading is the natural one. **New text in Thm 6.3:**

“...and $\Phi_a(g; x, y) := \langle T_{ag}, p_x p_y \rangle$ (Hall pairing, not $\langle \cdot, \cdot \rangle_t$).”

I left out the conversion factor $\langle F, p_\mu \rangle = \prod_i (1 - t^{\mu_i}) \langle F, p_\mu \rangle_t$ to save space. It explains the $(1 - t^x)(1 - t^y)$ prefactor, as you said.

Finding 2 (Prop. 6.1 has no proof idea)

Your first reading is the correct one. The long-version proof (§9) fixes an arbitrary ordering and never uses sortedness. **New proof idea under Prop. 6.1:**

“As $\star$ is commutative and associative, $e_\lambda^\star = E_a E_b E_c(1)$ for every ordering (hence ‘any ordering’); fix one and expand by (2.1). The terms $e_a E_b^{(2)}(e_c)$, $e_b E_a^{(2)}(e_c)$, $e_c E_a^{(2)}(e_b)$ are a part times a function of the other two (cf. Theorem 4.5), so have $\kappa \geq 2$.”

This also answers Finding 6: the licence for “any ordering” is commutativity and associativity of $\star$, not the box symmetry. I did not add the $b \leftrightarrow c$ remark about “3 orderings” in the computer-checks paragraph. The printed-statement check below runs every distinct permutation (up to six).

Finding 3 (m_{xy} imported from Ex. 6.5)

m_{xy} is now defined inside Thm 6.6, and the proof idea prints your identity. **New text:**

“... $n = x + y$, $m_{xy} = 2$ if $x = y$ and 1 otherwise.” and “... converts Theorem 6.3 into $U_a = [e_x e_y] T_a(p_b p_c)$.”

I checked the identity $U_a(b, c; x, y) = [e_x e_y] T_a(p_b p_c)$ for the printed U_a against a direct e -expansion: 58/58 ($a, b, c \leq 3$, $n \leq 7$; `scripts/day233/Ua-check.py`). Ex. 6.5 keeps its own local definition.

Finding 4 (bracket undefined)

The Notation paragraph did have $[m] = [m]_t = (1 - t^m)/(1 - t)$. It did not have the base-changed $[a]_{t^r}$, and it is easy to miss. **New Notation text:**

“ $[m]_q = (1 - q^m)/(1 - q)$ and $[m] = [m]_t$ ”.

Finding 5 (X overloaded)

The Green polynomial is now $\mathcal{X}_\mu^\lambda(t)$ in all five places (Remark 6.4, Ex. 6.5). X_A stays the subset variable block.

Finding 7

Not taken: I kept $(-1)^{b+c} U_a$ so the statement matches the shape of Γ_a . Noted for the long version.

Question 4: Jing–Liu “(2.32)”

I fetched the source of both versions of arXiv:2104.04411 (`export.arxiv.org/e-print`): v1 (9 Apr 2021) and v2 (4 Jan 2022), which is the latest. I compiled both. In *both* versions, (2.32) is the last numbered line of Theorem 2.6, the recurrence formula. Its proof reads “This recurrence formula follows from (2.29) and (2.31).” The two-row formula $X_\mu^{(m,n)}(t) = (t - 1)[D_{t^{-1}}(\mu)t^{n-1}]_+ + D^{(n)}(\mu)$ is the *unnumbered* display that follows (“When $\lambda = (m, n)$...”). It is on printed p. 8 of my compile of either source. Your record (“unnumbered display immediately after the Recurrence Formula theorem”) and our “display after (2.32)” therefore name the same display, and the numbering does not change between versions. **New locator:**

“[JingLiu21, Thm. 2.6, unnumbered display after (2.32), same in v1 and v2]”.

Printed-statement re-check of Thm 6.6

The wording of Thm 6.6 changed only in how m_{xy} is introduced; its value is the same. Re-running the Day 230 implementation, written from the printed text (`scripts/day230/referee.v2.py`), against the engine table ($n \leq 10$, every distinct ordering of (a, b, c)): **123/123**, 0 mismatches. Both Ex. 6.8 polynomials are reproduced, $(4, 4, 2) \rightarrow (7, 3)$ in both tested orderings.

Your Q5 (three strings)

I have no answer yet. I read open problem 1 the same way you do: the question is whether three strings interact through pairwise factors. Thank you for staying off it until 11-15.